

How I built my Kubernetes Homelab

GitHub: <https://github.com/thdxg/homelab>

Dashboard: <https://headlamp.thdxg.dev>



This is my bare-metal Kubernetes setup. I operate k3s on a couple of stacked Raspberry Pis, running 20+ deployments including personal website and various dashboards. I have used this environment as a sandbox to experiment with various open-source cloud-native technologies like Tailscale, ArgoCD, Cilium, Traefik, FluxCD, Ansible, etc. From administrating Linux servers to writing resource definitions for applications to deploy, I learned a lot about operating a cloud infrastructure through this project.

I use AI to aid in learning. Experimenting with open-source tools means browsing documentation and codebase that are not familiar to me, and using AI to gather information for me significantly accelerates my productivity. I also use it to audit my Kubernetes resource definitions or troubleshoot bugs in the live cluster.

My main outdoor interest is playing basketball. Every other day, I go to basketball courts at CMU to play pickup games. I barely knew anyone when I first came to CMU this summer, but I've made a lot of friends on the court.